Name

Mathematics Stage 5 Path

General and Point-Gradient Form of a Line

Book 1 Use various forms of the equation of a straight line

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Feedback: https://MrDingMaths.com

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Syllabus Content

MA5-LIN-P-01 describes and applies transformations, the midpoint, gradient/slope and distance formulas, and equations of lines to solve problems

Use various forms of the equation of a straight line

- Rearrange linear equations from gradient–intercept form $(y = mx + c)$ to general form $(ax + by + c = 0)$ and vice versa
- Find the x- and y-intercepts of a straight line in any form
- Graph the equation of a straight line in any form
- Use the point–gradient form $(y - y_1 = m(x - x_1))$ or the gradient–intercept form $(y = mx + c)$ to find the equation of a line passing through a point (x_1, y_1), with a given gradient m
- Use the gradient and the point–gradient form to find the equation of a line passing through 2 points
- Find the equation of a line that is parallel or perpendicular to a given line in any form
- Determine and justify whether 2 given lines are parallel or perpendicular

General Form

General Form of a Line

The general form of a line is:

$$ax + by + c = 0$$

Where a, b, c are expressed as integers if possible, and a is preferred to be positive.

Identify General form of a line

✗	✓	✓	✗
$y = 3x + 5$	$3x - y + 5 = 0$	$99x + 25y = 0$	$-99x + 25y = 0$

Identify the lines written in general form:

a	$y = 2x + 3$	**b**	$2x - y + 3 = 0$	**c**	$x - 3y + 6 = 0$
d	$-x + 3y - 6 = 0$	**e**	$0.5x + y - 2 = 0$	**f**	$3x - 4y - \frac{1}{5} = 0$

Example Rearrange an equation of a line to general form.

Rearrange $y = 3x - 7$ to general form.

$-3x + y + 7 = 0$

$3x - y - 7 = 0$

Guided Practice

Rearrange to general form.

a	$y = 5x - 2$	**b**	$y = \frac{1}{2}x - 3$
	$5x - y - 2 = 0$		$x - 2y - 6 = 0$
c	$y = 2x - \frac{3}{5}$	**d**	$y = \frac{x}{4} + \frac{1}{3}$
	$10x - 5y - 3 = 0$		$3x - 12y + 4 = 0$

1 Rearrange to general form $ax + by + c = 0$

a $y = x + 2$

$x - y + 2 = 0$

b $y = 3x - 1$

$3x - y - 1 = 0$

c $y = 8 + 5x$

$5x - y + 8 = 0$

d $x + 2y = 3$

$x + 2y - 3 = 0$

e $x - 2y = 6$

$x - 2y - 6 = 0$

f $y = 8x + 2$

$8x - y + 2 = 0$

g $y + 3 = 6x$

$6x - y - 3 = 0$

h $2y = x - 6$

$x - 2y - 6 = 0$

i $y = \frac{3x}{5} + 2$

$3x - 5y + 10 = 0$

2 Rearrange to general form $ax + by + c = 0$

a $y = \frac{x}{3} + \frac{1}{4}$

$4x - 12y + 3 = 0$

b $y = -\frac{5x}{8} + \frac{3}{2}$

$5x + 8y - 12 = 0$

c $y = \frac{3x}{4} - \frac{2}{5}$

$15x - 20y - 8 = 0$

d $y = -\frac{x}{2} - \frac{3}{4}$

$2x + 4y + 3 = 0$

e $y = \frac{3x}{2} + \frac{2}{5}$

$15x - 10y + 4 = 0$

f $y = -\frac{9x}{10} - \frac{7}{15}$

$27x + 30y + 14 = 0$

Sketching using Intercepts

ⓘ Finding the x and y intercepts

y intercept: substitute $x = 0$ and solve.

x intercept: substitute $y = 0$ and solve.

Example Identify x and y intercept from the equation

Find the x and y intercepts of

$$y = 4x + 2$$

y-int: $x = 0$

$y = 4(0) + 2$

$y = 2$

y-int is at $(0,2)$

x-int: $y = 0$

$0 = 4x + 2$

$-2 = 4x$

$-\frac{2}{4} = x$

$-0.5 = x$

x-int is at $(-0.5\ ,0)$

Find the x and y intercepts of

$$x + 2y - 6 = 0$$

y-int: $x = 0$

$0 + 2y - 6 = 0$

$2y = 6$

$y = 3$

y-int is at $(0,3)$

x-int: $y = 0$

$x + 2(0) - 6 = 0$

$x - 6 = 0$

$x = 6$

x-int is at $(6\ ,0)$

Guided Practice

Find the x and y intercepts of these lines.

a $y = 2x - 8$

y int: $x = 0$

$y = 2(0) - 8$

x int: $y = 0$

$0 = 2x - 8$

x-int: $(4,0)$, y-int: $(0,-8)$

b $x - y = 7$

y int: $x = 0$

x int: $y = 0$

x-int: $(7,0)$, y-int: $(0,-7)$

c $-3x - 2y = 6$

x-int: $(-2,0)$, y-int: $(0,-3)$

d $4x + 4y = 7$

x-int: $\left(\frac{7}{4},0\right)$, y-int: $\left(0,\frac{7}{4}\right)$

Sketching lines using the intercepts

1. Find the y intercept ($x = 0$)
2. Find the x intercept ($y = 0$)
3. Connect the x and y intercept and extend the straight line.

Example Sketch graphs from the equation by finding the intercepts

Sketch $x + 2y + 6 = 0$

1. y-int: $x = 0$

$2y + 6 = 0$

$2y = -6$

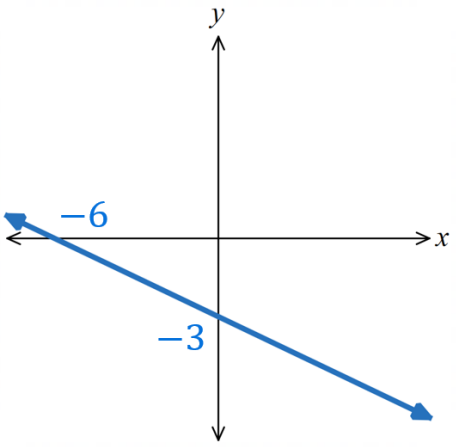
$y = -3$

2. x-int: $y = 0$

$x + 6 = 0$

$x = -6$

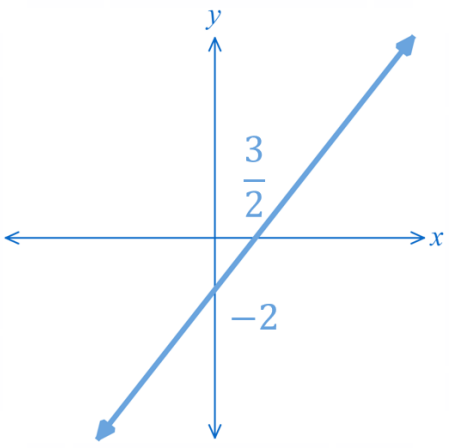
3.



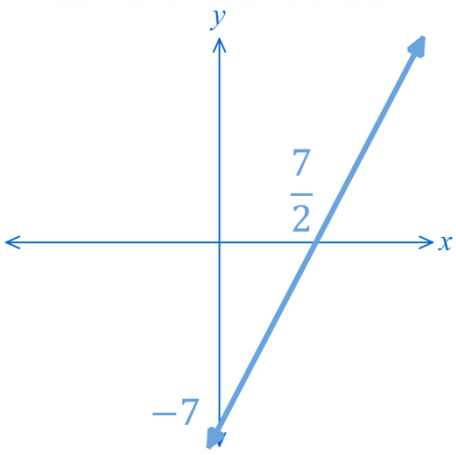
Guided Practice

Sketch by finding the intercepts:

a $3y = 4x - 6$



b $4x - 2y = 14$



1 Determine the x and y intercepts for these lines.

a $y = x + 1$

y int: $x = 0$ | x int: $y = 0$

x-int: (−1, 0), y-int: (0, 1)

b $y = \frac{1}{2}x - 3$

x-int: (6, 0), y-int: (0, −3)

c $y = x$

x-int: (0, 0), y-int: (0, 0)

d $y - 3x = 0$

x-int: (0, 0), y-int: (0, 0)

e $x + y - 1 = 0$

x-int: (1, 0), y-int: (0, 1)

f $x - 3y - 3 = 0$

x-int: (3, 0), y-int: (0, −1)

g $2x - 3y - 12 = 0$

x-int: (6, 0), y-int: (0, −4)

h $5x + 2y - 10 = 0$

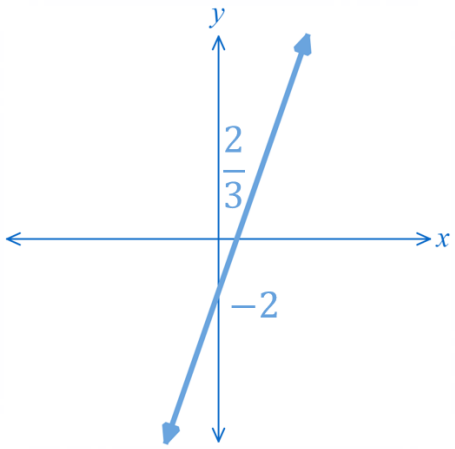
x-int: (2, 0), y-int: (0, 5)

2 Follow these steps for the equation $y = 3x - 2$

a Find the y-intercept by substituting $x = 0$.

b Find the x-intercept by substituting $y = 0$ then solving.

c Sketch the graph on the axis.

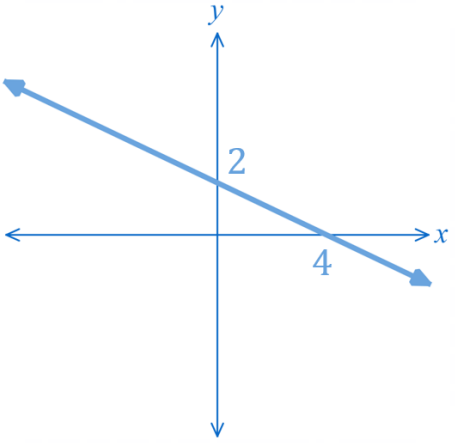


3 Follow these steps for the equation $x + 2y - 4 = 0$

a Find the y-intercept by substituting $x = 0$.

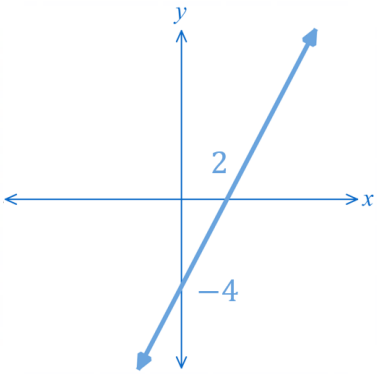
b Find the x-intercept by substituting $y = 0$ then solving.

c Sketch the graph on the axis.

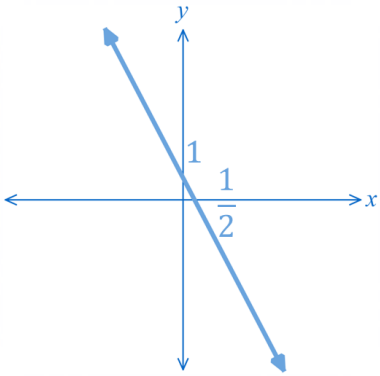


4 Sketch these lines. Show the y-intercept and x-intercept.

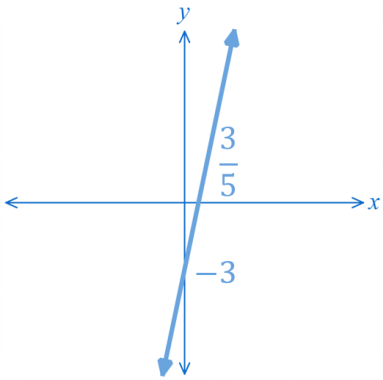
a $y = 2x - 4$



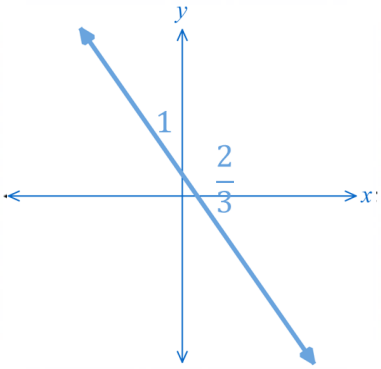
b $y = -2x + 1$



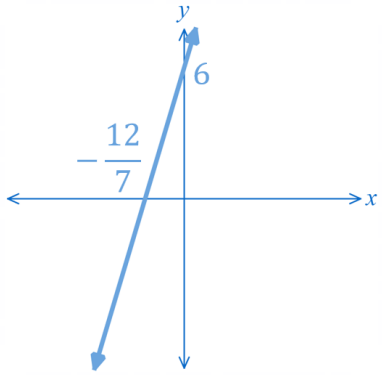
c $y = 5x - 3$



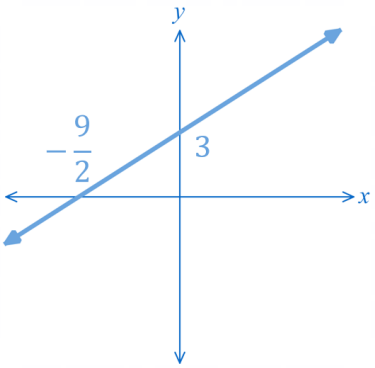
d $y = -\frac{3}{2}x + 1$



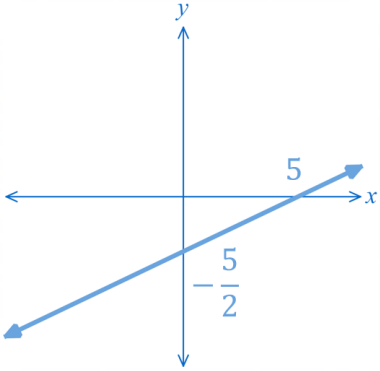
e $y = \frac{7}{2}x + 6$



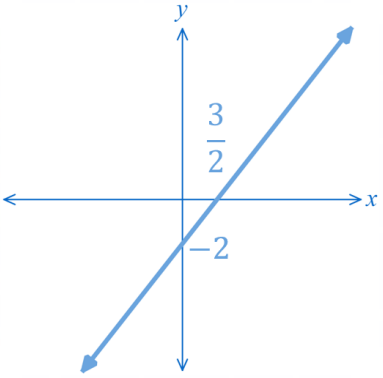
f $y = 3 + \frac{2}{3}x$



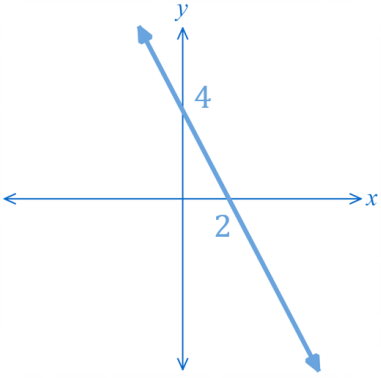
g $y = \frac{1}{2}x - 2.5$



h $y = \frac{4x}{3} - 2$

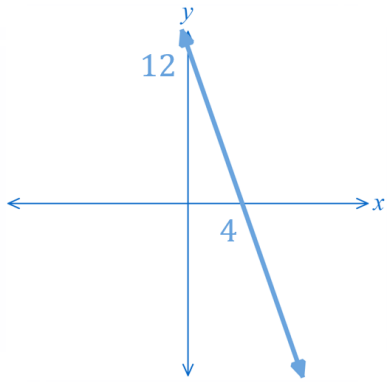


i $y = 4 - 2x$

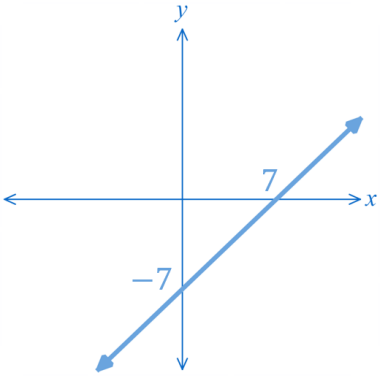


5 Sketch these graphs by finding the x- and y-intercepts.

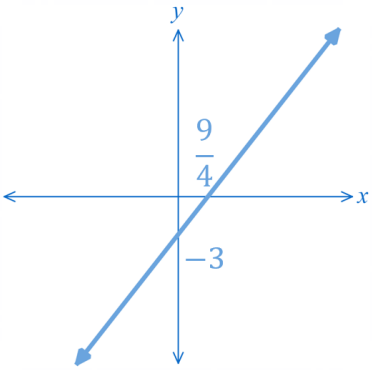
a $3x + y = 12$



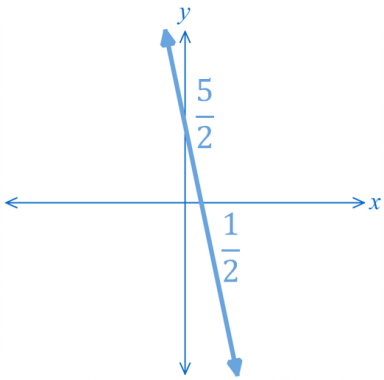
b $x - y = 7$



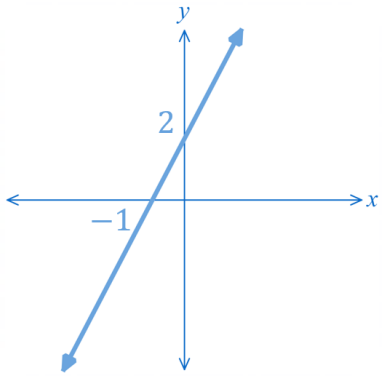
c $4x - 3y = 9$



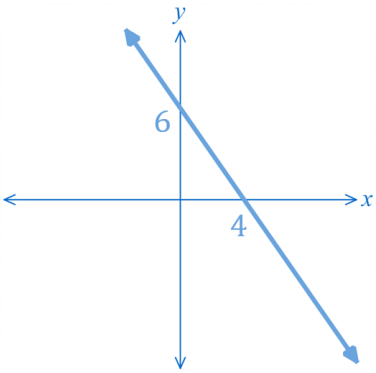
d $10x + 2y = 5$



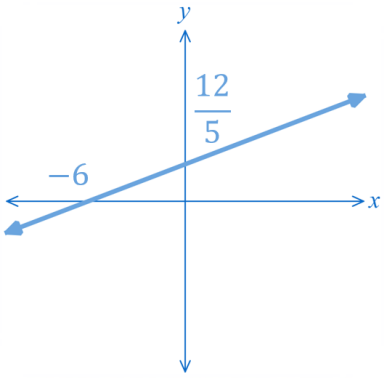
e $2x - y + 2 = 0$



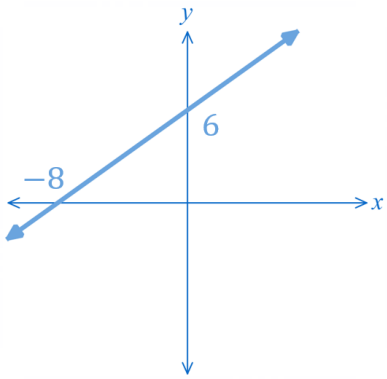
f $3x + 2y = 12$



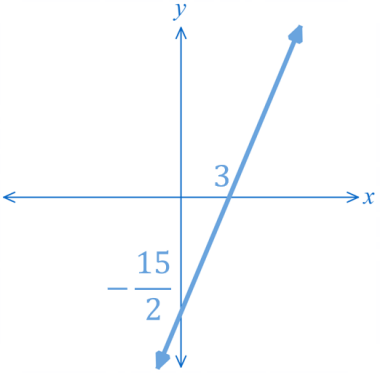
g $5y - 2x = 12$



h $4y - 3x = 24$



i $-5x + 2y + 15 = 0$



Finding the Equation given Gradient and a Point

ⓘ Point-Gradient Formula

The point-gradient formula of a line relates the gradient m and any point on the line (x_1, y_1)

$$y - y_1 = m(x - x_1)$$

- m: gradient
- y_1: y coordinate of a point
- x_1: x coordinate of a point

Example Find equation of a line given gradient and a point

Find the equation of the line with gradient 3 passing through $(6,4)$.

m, $x_1\ y_1$

$$y - y_1 = m(x - x_1)$$
$$y - 4 = 3(x - 6)$$
$$y - 4 = 3x - 18$$
$$y = 3x - 14$$

Find the equation of the line with gradient -4 and x intercept 1.

m, $(x_1\ y_1) = (1,0)$

$$y - y_1 = m(x - x_1)$$
$$y - 0 = -4(x - 1)$$
$$y = -4x + 4$$

Guided Practice

Find the equation of the line, in gradient-intercept form, with:

a Gradient 4 and passes through $(-1,2)$

$y = 4x + 6$

b Gradient 3 and passes through $(-3,5)$

$y = 3x + 14$

c Gradient $\frac{1}{3}$ and passes through $(9,2)$

$y = \frac{x}{3} - 1$

d Gradient -3 passes through $(8,-5)$

$y = -3x + \frac{13}{3}$

1 Use $y - y_1 = m(x - x_1)$ to find the equation of these lines in gradient-intercept form.

a $(3,2)$ and $m = 4$

$y - y_1 = m(x - x_1)$

$y - 2 = 4(x - 3)$

$y = 4x - 10$

b $(4, -1)$ and $m = -1$

$y = -x + 3$

c $(-7,2)$ and $m = \frac{1}{2}$

$y = \frac{x}{2} + \frac{11}{2}$

d $(2,5)$ and $m = 2$

$y = 2x + 1$

e $(-6,4)$ and $m = -1$

$y = -x - 2$

f $(3, -8)$ and $m = 4$

$y = 4x - 20$

2 Find the equation of the line:

a through (1,1) with gradient 2

$y = 2x - 1$

b with gradient −1 through (3, 1)

$y = -x + 4$

c through (0,0) with gradient −5

$y = -5x$

d through (−1,3) with gradient $-\frac{1}{3}$

$y = -\frac{x}{3} + \frac{8}{3}$

e with gradient $-\frac{4}{5}$ through (3, −4).

$y = -\frac{4x}{5} - \frac{8}{5}$

f through $(-5,3)$ with gradient $-\frac{3}{5}$

$y = -\frac{3x}{5}$

3 A line passes through the y-axis at (0, 6) and has a gradient of $\frac{5}{7}$. What is its equation?

$y = \frac{5x}{7} + 6$

4 Find the equation of a line with a gradient of -4 and an x-intercept of 5.

$y = -4x + 20$

Finding the Equation given Two Points

ⓘ Finding Equation of a Line given Two Points

1. Calculate gradient using the formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$
2. Substitute into point-gradient form. $y - y_1 = m(x - x_1)$
3. Rearrange to $y = mx + c$.

Example Find equation of a line given two points

Find the equation of the line, in gradient-intercept form, passing through $(1, 5)$ and $(4, -1)$

1. $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-5}{4-1}$

$= -2$

2. Sub $m = -2$ and (1,5)

$y - y_1 = m(x - x_1)$

$y - 5 = -2(x - 1)$

3. $y - 5 = -2x + 2$

$y = -2x + 7$

Guided Practice

Find the equation of the line, in gradient-intercept form, passing through:

a $(2,7)$ and $(-1.5,0)$

1. $m = \frac{y_2 - y_1}{x_2 - x_1}$

2. $y - y_1 = m(x - x_1)$

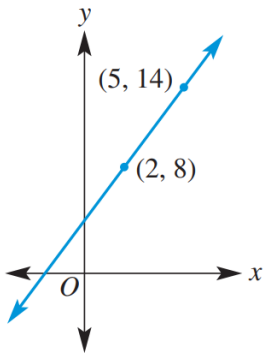
3.

$y = 2x + 3$

b $(1,7)$ and $(3, -1)$

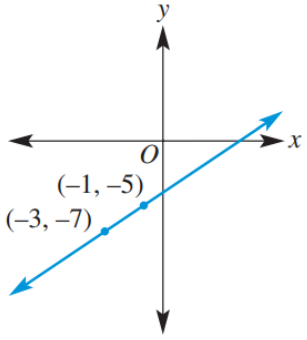
$y = -4x + 11$

c



$y = 2x + 4$

d



$y = x - 4$

1 Determine the equation of the line that pass through these two points.

a $(4,2)$ and $(12,4)$

$y = \frac{x}{4} + 1$

b $(1,4)$ and $(3,8)$

$y = 2x + 2$

c $(-1,3)$ and $(2,0)$

$y = -x + 2$

d $(7,3)$ and $(10,6)$

$y = x - 4$

e $(2,-2)$ and $(-1,6)$

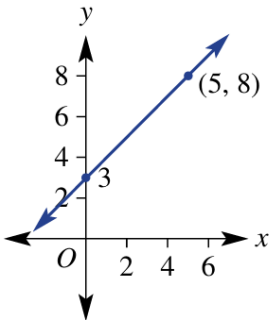
$y = -\frac{8x}{3} + \frac{10}{3}$

f $(-10,2)$ and $(1,-4)$

$y = -\frac{6x}{11} - \frac{38}{11}$

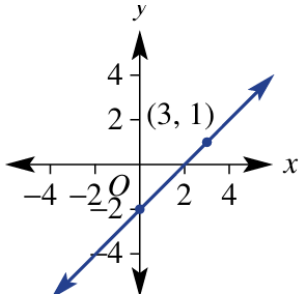
2 Find the equation of these lines using an appropriate method.
(Hint: some of these graphs already give you the y-intercept, so you only need the gradient)

a



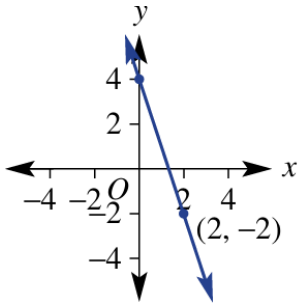
$y = x + 3$

b



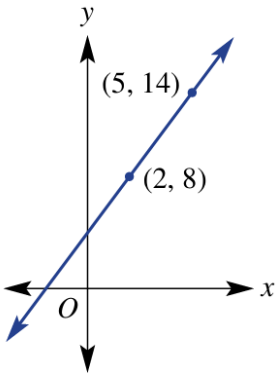
$y = x - 2$

c



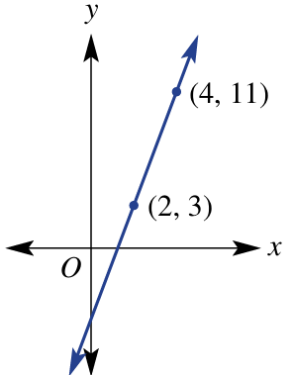
$y = -3x + 4$

d



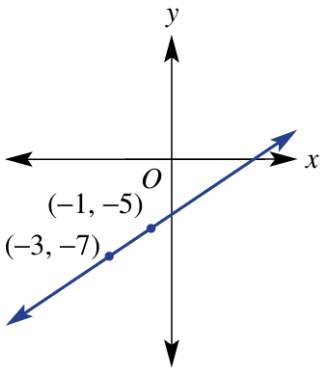
$y = 2x + 4$

e



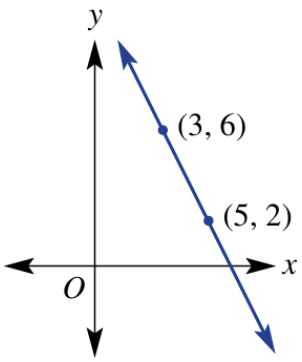
$y = 4x - 5$

f



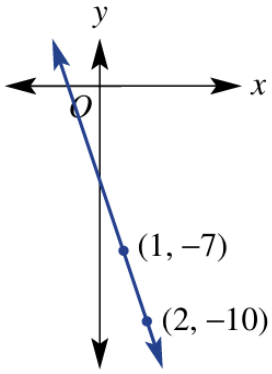
$y = x - 4$

g



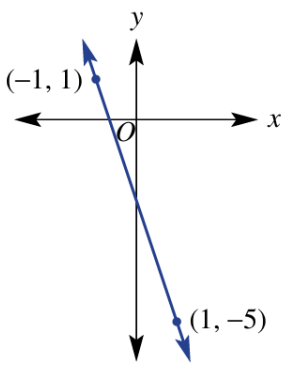
$y = -2x + 12$

h



$y = -3x - 4$

i



$y = -3x - 2$

3 Consider the points P(4, 7) and Q(6, −1).

a Calculate the length of PQ.

$\sqrt{68}$

b Find the midpoint of the PQ.

(5, 3)

c What is the gradient of PQ?

-4

d Find the equation of the line passing through P and Q.

$y = -4x + 23$

4 A line with gradient $\frac{2}{3}$ passes through the midpoint of $(5,6)$ and $(1,10)$.
Find the equation of the line.

Midpoint: (3, 8)

$y = \frac{2x}{3} + 6$

5 A line with gradient $-\frac{3}{5}$ passes through the midpoint of $(-8,2)$ and $(-2,20)$.
Find the equation of the line.

Midpoint: $(-5,11)$

$y = -\frac{3x}{5} + 8$

6 Determine the equation of a line with gradient -4 and passes through $(3, -10)$.

Method 1: Substitute into $y - y_1 = m(x - x_1)$

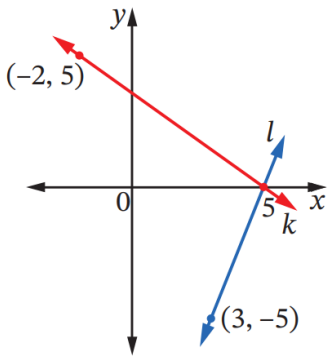
$y + 10 = -4(x - 3)$
$y = -4x + 2$

Method 2: Substitute into $y = mx + c$ and solve for c

$-10 = -4(3) + c$
$c = 2$
$y = -4x + 2$

7 The diagram shows line k and line l. Find:

a The equation of line l in general form.



$5x - 2y - 25 = 0$

b The equation of line k in general form.

$5x + 7y - 25 = 0$

c w if the point $(7, w)$ lies on l.

$w = 5$

d t if the point $(t, 11)$ lies on k.

$t = -\frac{52}{5}$

8

a A line with gradient 4 passes through $(1, 5)$ and $(-2, k)$. Find the value of k.

$k = -7$

b Find the value of a if $2x + 5y - 6 = 0$ passes through $(a, -2)$.

$a = 8$

c Find the value of a if $ax - 8y + 3 = 0$ passes through $(-7, 3)$.

$a = -3$

d Find the value(s) of a if $ax + y = 6$ passes through $(a, 5a)$

$a = -6$ or $a = 1$

9 This is the **intercept form of a straight line**.

$$\frac{x}{a}+\frac{y}{b}=1$$

a Find the x and y-intercepts of this line and hence explain why it's called the intercept from of a straight line.

When $y = 0: x = a$ (x-intercept is a)

When $x = 0: y = b$ (y-intercept is b)

It's called the intercept form because a and b directly give the x and y-intercepts.

b Use the intercept form to find the equation of a line with x-intercept at -2 and y-intercept at 3.

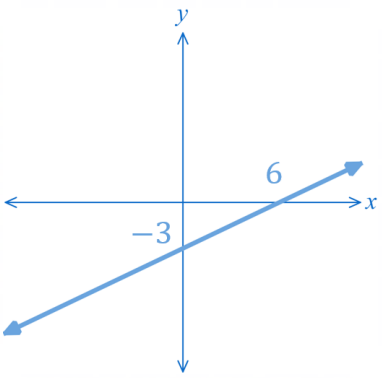
$\frac{x}{-2}+\frac{y}{3}=1$

$3x - 2y = -6$

c Use this knowledge to quickly sketch these lines without any calculations.

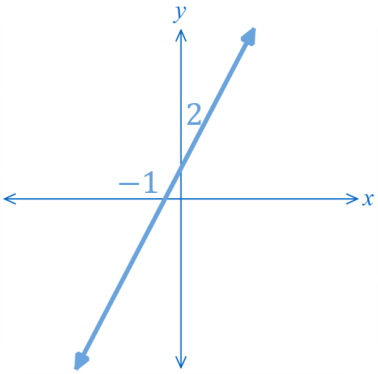
i

$$\frac{x}{6}-\frac{y}{3}=1$$



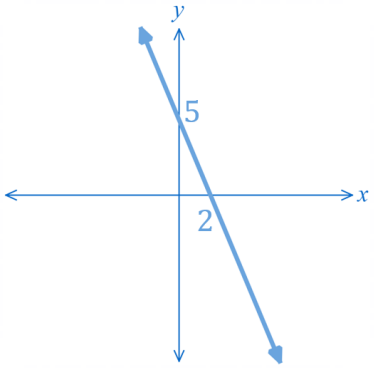
ii

$$-x+\frac{y}{2}=1$$



iii

$$\frac{x}{2}+\frac{y}{5}=1$$



10 Another form of a line is the **two-point form** that states the equation of a line that passes through any two points $(x_1 y_1)$ and (x_2, y_2). We will derive it in this section.

a What is the expression for the gradient between the two points $(x_1 y_1)$ and (x_2, y_2)?

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

b Use the formula $y - y_1 = m(x - x_1)$ and the expression in part **a** to show that the two-point form of a line is $\frac{y-y_1}{x-x_1} = \frac{y_2-y_1}{x_2-x_1}$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

c Use the two-point form to find the equation of a line passing through $(7,3)$ and $(10,6)$. Answer in gradient-intercept form. Check your answer using DESMOS.

$$\frac{y-3}{x-7} = \frac{6-3}{10-7}$$

$$\frac{y-3}{x-7} = 1$$

$$y - 3 = x - 7$$

$$y = x - 4$$

d Find the equation of a line passing through $(-1,10)$ and $(10, -17.5)$. Answer in gradient-intercept form. Check your answer using DESMOS.

$$\frac{y-10}{x+1} = \frac{-17.5-10}{10+1}$$

$$\frac{y-10}{x+1} = -2.5$$

$$y - 10 = -2.5(x + 1)$$

$$y = -2.5x + 7.5$$

Finding the Equation of Parallel and Perpendicular Lines

Review

☐ Recall parallel and perpendicular gradients

a Gradient parallel to $m = 3$

b Gradient parallel to $m = \frac{2}{3}$

c Gradient perpendicular to $m = \frac{2}{3}$

d Gradient perpendicular to $m = -5$

e Gradient parallel to $y = \frac{x}{7} + 3$

f Gradient perpendicular to $y = \frac{3x}{2} - 4$

☐ Find equation of parallel/vertical line, given y-intercept of new line

a Find the equation of the line parallel to $y = 5x - 2$ passing through $(0,1)$

b Find the equation of the line perpendicular to $y = \frac{3x}{7} - 2$ passing through $(0, -5)$

☐ Find equation of line given gradient and a point

a Find the equation of the line with gradient 3 passing through $(4,5)$

b Find the equation of the line with gradient $-\frac{3}{5}$ passing through $(2, -5)$

Finding the Equation of a Parallel and Perpendicular Line

Gradient of a Parallel Line

$$m_{\text{par}} = m_{\text{original}}$$

Gradient of Perpendicular Line

$$m_{\text{perp}} = -\frac{1}{m_{\text{original}}}$$

1. Determine gradient m_{new} of the parallel/perpendicular line.
2. Write the equation $y - y_1 = m_{\text{new}}(x - x_1)$
3. Substitute new gradient and given point into the formula.
4. Rearrange to $y = mx + c$

Example Find the equation of a parallel/perpendicular line

Find the equation of a line, in gradient-intercept form, that is:

Parallel to $y = -2x - 7$ through $(1,9)$

1. $m_{\text{new}} = -2$
2. $y - y_1 = m_{\text{new}}(x - x_1)$
3. $y - 9 = -2(x - 1)$
4. $y - 9 = -2x + 2$

   $y = -2x + 11$

Perpendicular to $y = \frac{3}{4}x - 1$ through $(3, -2)$

1. $m_{\text{new}} = -\frac{4}{3}$
2. $y - y_1 = m_{\text{new}}(x - x_1)$
3. $y - (-2) = -\frac{4}{3}(x - 3)$
4. $y + 2 = -\frac{4x}{3} + 4$

   $y = -\frac{4x}{3} + 2$

Guided Practice

Find the equation of the line, in gradient-intercept form that is:

a Parallel to $y = x + 3$ and passes through $(1, 5)$

1. $m_{\text{new}} =$ .....
2. $y - y_1 = m_{\text{new}}(x - x_1)$
3.
4.

$y = x + 4$

b Parallel to $y = \frac{2}{3}x + 1$ and passes through $(3, -4)$

1. $m_{\text{new}} =$ .....
2. $y - y_1 = m_{\text{new}}(x - x_1)$
3.
4.

$y = \frac{2x}{3} - 6$

c Perpendicular to $y = 2x + 3$ and passes through $(2, 5)$

$y = -\frac{x}{2} + 6$

d Perpendicular to $y = -4x + 1$ and passes through $(-4, -3)$

$y = \frac{x}{4} - 2$

1 Find the gradient of a line that is parallel to each line.

a $y = 4x + 6$

4

b $y = -7x - 1$

-7

c $y = -\frac{3}{4}x + 2$

$-\frac{3}{4}$

d $y = \frac{8x}{7} - \frac{1}{2}$

$\frac{8}{7}$

2 Find the gradient of a line that is perpendicular to each line.

a $y = 3x - 1$

$-\frac{1}{3}$

b $y = -2x + 6$

$\frac{1}{2}$

c $y = \frac{7}{8}x - \frac{2}{3}$

$-\frac{8}{7}$

d $y = \frac{-4x}{9} - \frac{4}{7}$

$\frac{9}{4}$

3 Decide if these pairs of lines are parallel, perpendicular, or neither.

a $y = 3x - 1$
$y = 3x + 7$

parallel

b $y = \frac{1}{2}x - 6$
$y = \frac{x}{2} - 4$

parallel

c $y = -\frac{2}{3}x + 1$
$y = \frac{2}{3}x - 3$

neither

d $y = -8x + 4$
$y = \frac{1}{8}x - 2$

perpendicular

e $x - y = 4$
$y = -x + \frac{1}{2}$

perpendicular

f $3x - y = 2$
$x + 3y = 5$

perpendicular

4 Find the equation of:

a the line through (2, 5) and parallel to $y = 2x + 5$

$y = 2x + 1$

b the line through (2, 5) and perpendicular to $y = 2x + 5$

$y = -\frac{x}{2} + 6$

c the line through (−7, 6) and parallel to $y = \frac{3x}{7} - 8$

$y - \frac{3x}{7} + 9$

d the line through (−4, 0) and perpendicular to y = $-\frac{2}{5}x$

$y = \frac{5x}{2} + 10$

5 Write down the line that is parallel to the given lines and passes through the given point.

Parallel to	Goes through	Answer
$y = 3x + 2$	$(0,7)$	$y = 3x + 7$
$y = 3x + 2$	$(3,0)$	$y = 3x - 9$
$y = 3x + 2$	$(3,7)$	$y = 3x - 2$
$y = 3x + 10$	$(3,7)$	$y = 3x - 2$
$y = 2x - 2$	$(3,7)$	$y = 2x + 1$
$y = 2x - 2$	$(-3,7)$	$y = 2x + 13$
$y = 2x - 2$	$(0,0)$	$y = 2x$
$y = -3$	$(8,11)$	$y = 11$

6 Write down the line that is perpendicular to the given lines and passes through the given point.

Perpendicular to	Goes through	Answer
$y = -x + 2$	$(0,7)$	$y = x + 7$
$y = 2x + 2$	$(0,7)$	$y = -\frac{x}{2} + 7$
$y = 2x + 2$	$(12,7)$	$y = -\frac{x}{2} + 13$
$y = 3x + 2$	$(12,7)$	$y = -\frac{x}{3} + 11$
$y = 3x + 2$	$(10,7)$	$y = -\frac{x}{3} + \frac{31}{3}$
$y = 3x + 2$	$(-12,7)$	$y = -\frac{x}{3} + 3$
$y = 2x + 2$	$(0,0)$	$y = -\frac{x}{2}$
$x = 7$	$(0,3)$	$y = 3$

7 Find the equation of the line that is:

a Perpendicular to $y = \frac{2x-1}{3}$ passing through $(0,5)$

$y = -\frac{3x}{2} + 5$

b Parallel to $y = \frac{3-5x}{7}$ passing through $(1,7)$

$y = -\frac{5x}{7} + \frac{54}{7}$

c Parallel to $3y - 2x = 3$ passing through $(-2,4)$

$y = \frac{2x}{3} + \frac{16}{3}$

d Perpendicular to $7x - y = -1$ passing through $(-3,-1)$

$y = -\frac{x}{7} - \frac{10}{7}$

e Parallel to $y = -\frac{3}{7}$ passing through $\left(\frac{2}{3}, \frac{1}{2}\right)$

$y = \frac{1}{2}$

f Perpendicular to $3x + 4y - 3 = 0$ passing through $(-2, 1)$.

$y = \frac{4x}{3} + \frac{11}{3}$

g parallel to the x-axis passing through $(-2,6)$.

$y = 6$

h parallel to the y-axis passing through $(3, 5)$.

$x = 3$

8 Find the value of a if:

a $y = \frac{a}{7}x + b$ is parallel to $y = 2x - 4$

$\frac{a}{7} = 2$

$a = 14$

b $y = \frac{2a+1}{3}x + b$ is parallel to $y = -x - 3$

$\frac{2a+1}{3} = -1$

$2a + 1 = -3$

$a = -2$

c $y = \left(\frac{1-a}{2}\right)x + b$ is perpendicular to $y = \frac{1}{2}x - \frac{3}{5}$

$\left(\frac{1-a}{2}\right)\left(\frac{1}{2}\right) = -1$

$\frac{1-a}{4} = -1$

$1 - a = -4$

$a = 5$

d $ay = 3x + b$ is perpendicular to $y = -\frac{3}{7}x - 1$

$y = \frac{3}{a}x + \frac{b}{a}$

$\left(\frac{3}{a}\right)\left(-\frac{3}{7}\right) = -1$

$-\frac{9}{7a} = -1$

$9 = 7a$

$a = \frac{9}{7}$

Explain why your answers above don't depend on the value of b.

Parallel and perpendicular depends only on the gradient.

9 The line $3x - 8y + 10 = 0$ and another line intersect at right angles at the point $(10, 5)$.
Find the equation of the other line.

Perpendicular gradient: $-\frac{8}{3}$

$y - 5 = -\frac{8}{3}(x - 10)$

$3(y - 5) = -8(x - 10)$

$3y - 15 = -8x + 80$

$8x + 3y = 95$

$y = \frac{-8x + 95}{3}$

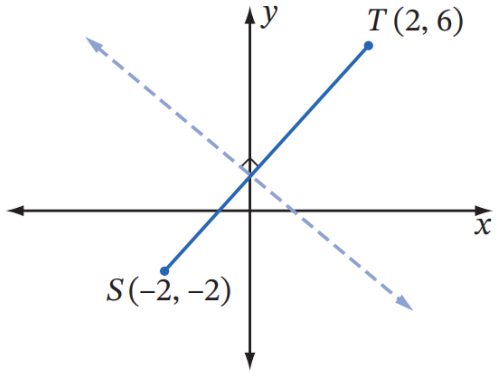
10 In this question, you will find the *perpendicular bisector* of an interval.

a Find the gradient of interval ST

$m = 2$

b Find the midpoint of ST

$M(0,2)$



c The dotted line is perpendicular to ST and passes through its midpoint. What is its gradient?

$-\frac{1}{2}$

d Find the equation of the dotted line.
Answer in gradient-intercept form $y = mx + c$.

$y = -\frac{1}{2}x + 2$

11

a Find the equation of the perpendicular bisector of the line segment joining $(1,1)$ with $(3,5)$ in general form.

Midpoint is $(2,3)$

Gradient is 2

Perp gradient $-\frac{1}{2}$

$x + 2y - 8 = 0$

b Find where this bisector cuts the x-axis.

$(8,0)$

12 The line $ax + by + 1 = 0$ is perpendicular to $4x + 3y = 7$ and passes through $(9,7)$.
Find the values of a and b.

Gradient of original line: $-\frac{4}{3}$

Perp gradient: $\frac{3}{4}$

$y - 7 = \frac{3}{4}(x - 9)$

$4y - 28 = 3x - 27$

$3x - 4y - 1 = 0$

$a = 3, b = -4$

13 The interval PQ has gradient −3. A second line passes through $A(-2,4)$ and $B(1,k)$.

a Find k if AB is parallel to PQ.

$m_{PQ} = -3$

For AB parallel to PQ: $\frac{k-4}{1-(-2)} = -3$

$\frac{k-4}{3} = -3$

$k = -5$

b Find k if AB is perpendicular to PQ.

For AB perpendicular to PQ: $\frac{k-4}{3} = \frac{1}{3}$

$k - 4 = 1$

$k = 5$

14 The line l with the equation $x - 2y + 4 = 0$ passes through the point $A(8,6)$ and intersects the x-axis at B as shown on the diagram below.
The line through A and perpendicular to l intersects the x-axis at C.

a Find the equation of AC in general form.

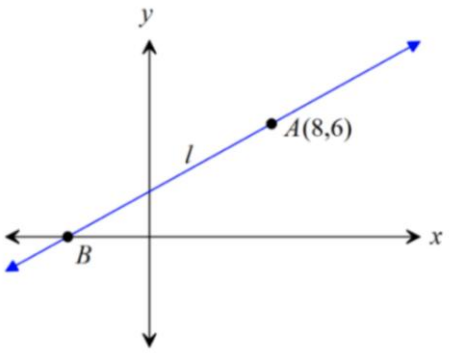
Gradient of l from $x - 2y + 4 = 0$ is $\frac{1}{2}$

Perpendicular gradient: -2

Through $A(8,6)$: $y - 6 = -2(x - 8)$

$y - 6 = -2x + 16$

$2x + y - 22 = 0$



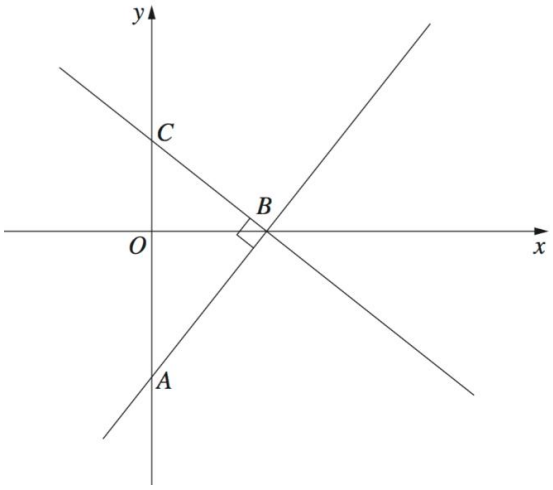
b Find the coordinates of C.

Let $y = 0$

$(11,0)$

15 The points A and C lie on the y-axis and the point B lies on the x-axis.
The line AB has equation $5x - 3y = 15$.
The line BC is perpendicular to AB.

a Find the coordinates of B.



Line AB: $5x - 3y = 15$

On x axis $(y = 0)$: $5x = 15, x = 3$

$B = (3, 0)$

b Find the equation of the line BC in general form.

Gradient of AB: $\frac{5}{3}$

Perpendicular gradient: $-\frac{3}{5}$

Through $B(3,0)$: $y - 0 = -\frac{3}{5}(x - 3)$

$5y = -3x + 9$

$3x + 5y - 9 = 0$

16 Line A has equation $2x + ky = 5$.
Line B passes through the points $(1,5)$ and $(4 + k, 6)$.
Find the value of k if the line A and B do not intersect.

Line A: $2x + ky = 5$, gradient $-\frac{2}{k}$

Line B through $(1,5)$ and $(4 + k, 6)$: gradient $= \frac{6-5}{4+k-1} = \frac{1}{3+k}$

For no intersection (parallel): $-\frac{2}{k} = \frac{1}{3+k}$

$-2(3 + k) = k$

$-6 - 2k = k$

$k = -2$

17 Given the points $X(-1, 0), Y(1, a)$ and $Z(a, 2)$, find a if $\angle YXZ = 90°$.

$XY \perp XZ$, so $m_{XY} \times m_{XZ} = -1$

$m_{XY} = \frac{a-0}{1-(-1)} = \frac{a}{2}$

$m_{XZ} = \frac{2-0}{a-(-1)} = \frac{2}{a+1}$

$m_{XY} \times m_{XZ} = -1$

$\frac{a}{a+1} = -1$

$a = -a - 1$

$a = -\frac{1}{2}$

18 For the four points $P(k,1), Q(-2,-3), R(2,3)$ and $S(1,k)$, it is known that PQ is parallel to RS. Find the possible values of k.

$$m_{PQ} = \frac{-3-1}{-2-k} = -\frac{4}{-2-k} = \frac{4}{2+k}$$

$$m_{RS} = \frac{k-3}{1-2} = \frac{k-3}{-1} = 3-k$$

$$\text{For parallel: } \frac{4}{2+k} = 3-k$$

$$4 = (3-k)(2+k)$$

$$4 = 6 + 3k - 2k - k^2$$

$$k^2 - k - 2 = 0$$

$$(k-2)(k+1) = 0$$

$$k = 2 \text{ or } k = -1$$

19 **2014 HSC Advanced Band 4**

Which equation represents the line perpendicular to $2x - 3y = 8$ passing through $(2,0)$?

A. $3x + 2y = 4$

B. $3x + 2y = 6$

C. $3x - 2y = -4$

D. $3x - 2y = 6$

$$\text{Gradient of } 2x - 3y = 8 \text{ is } \frac{2}{3}, \quad \text{Perpendicular gradient: } -\frac{3}{2}$$

$$\text{Through } (2,0): y = -\frac{3}{2}(x-2)$$

$$2y = -3x + 6$$

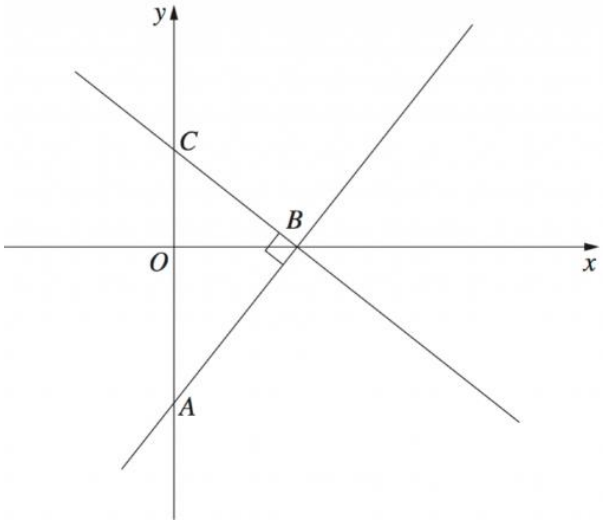
$$3x + 2y = 6$$

Answer: B

20 2009 HSC Advanced Band 4

In the diagram, A and C lie on the y-axis and the point B lies on the x-axis.
The line AB has equation $y = \sqrt{3}x - 3$. The line BC is perpendicular to AB.

a Find the equation of line BC in the form $y = mx + c$.



Gradient of AB: $\sqrt{3}$ Perpendicular gradient: $-\frac{1}{\sqrt{3}}$

B is x intercept of $y = \sqrt{3}x - 3$

$0 = \sqrt{3}x - 3$

$x = \sqrt{3}$

$B = (\sqrt{3}, 0)$

BC through B with gradient $-\frac{1}{\sqrt{3}}$: $y - 0 = -\frac{1}{\sqrt{3}}(x - \sqrt{3})$

$y\sqrt{3} = -x + \sqrt{3}$

$x + y\sqrt{3} = \sqrt{3}$

$y = -\frac{1}{\sqrt{3}}x + 1$

b Find the area of triangle ABC.

A is y-intercept of AB: x = 0 gives y = -3

C is y-intercept of BC: x = 0 gives y = 1

Base AC = 4 units, height = $\sqrt{3}$ (horizontal distance from y-axis to B)

Area $= \frac{1}{2}(4)(\sqrt{3}) = 2\sqrt{3}$ units^2

$2\sqrt{3}$ units^2